2024 15th International Conference on Information and Communication Technology Convergence (ICTC) | 979-8-3503-6463-7/24/$31.00 ©2024 IEEE | DOI: 10.1109/ICTC62082.2024.10827089

Trends in Transfer Learning on Reinforcement Learning for Mobility Control

Sungjoon Lee†, Soohyun Park§, and Joongheon Kim†
†Department of Electrical and Computer Engineering, Korea University, Seoul, Republic of Korea
§Division of Computer Science, Sookmyung Women's University, Seoul, Republic of Korea
E-mail: ssungjoon@korea.ac.kr, soohyun.park@sookmyung.ac.kr, joongheon@korea.ac.kr

***Abstract*—This paper provides an in-depth analysis of recent trends and advancements in applying Transfer Learning (TL) techniques to RL for vehicle control. We particularly focus on applications in autonomous driving and unmanned systems. The integration of TL with RL has demonstrated significant potential in overcoming key challenges such as Sim2Real transfer, multi-agent collaboration, and scalability in real-time applications.**

I. Introduction

As modern society demands more sophisticated and interconnected transportation systems, the integration of computing devices into automobiles has significantly advanced. Particularly, with the development of artificial intelligence, computing devices and algorithms embedded in vehicle control systems have become increasingly sophisticated. Utilizing datasets collected from driving environments, the advent of deep learning has spurred research on vehicle control using supervised, unsupervised, and reinforcement learning (RL) techniques. Many autonomous systems have applied RL to overcome spatial constraints, enabling adaptive control in dynamic situations. Consequently, research has focused on applying artificial intelligence to control of vehicles operating on land, in the air, ocean, and in space, allowing them to autonomously act and make decisions within predefined environments. Recent studies on land-based vehicles, which are easier to apply in real-world scenarios and are already commercialized, have focused on areas such as lane changing [1], vehicle control based on ethical decision-making [2], ensuring safety for rapid decision-making and speed control [3], embedding RL models that apply sophisticated dynamics into real-world systems [4], and vehicle control considering electric vehicle charging [5], [6]. In the field of aerial vehicles, research has predominantly focused on multi-agent reinforcement learning (MARL) control using multiple UAVs and UAV-based communication networks [7], [8]. In ocean environments, research has concentrated on robust underwater agent control considering the harsh conditions, while in space, studies have explored descent control for planetary exploration in complex six-degree-of-freedom environments with fuel optimization [9]. However, traditional RL begins training in new environments or tasks without prior knowledge, requiring extensive exploration and interaction samples to establish an initial policy, which can be time-consuming. This challenge makes it difficult to ensure real-time performance and safety in complex environments like autonomous driving systems. Transfer learning on RL (TRL) addresses this issue by significantly accelerating learning speed by applying knowledge gained from previously learned tasks to new, related tasks. This can be accomplished through various methods, such as using meta-learning techniques similar to transfer learning, or by applying the policies, value functions, or features learned from a previous task to a new task [10]. This approach allows agents to adapt more quickly to new tasks without the need to learn from scratch, thereby reducing training time, and improving performance compared to conventional RL algorithms.

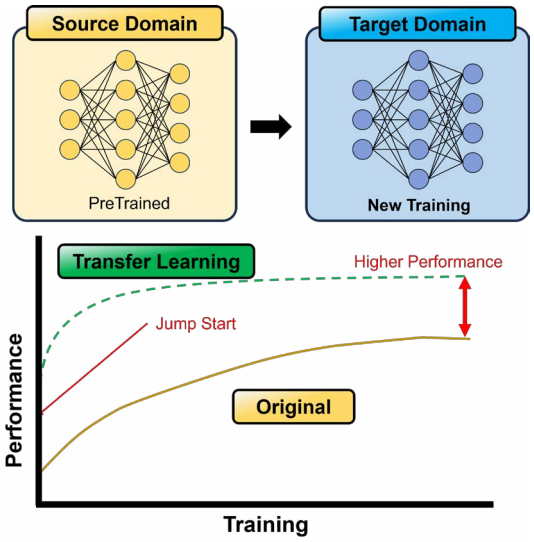

Fig. 1: The Advantages of Transfer Learning.

This paper will introduce the overall concept of TRL methods and review the literature that has applied these methods to vehicle control. The rest of the paper is organized as follows. Sec. II illustrates an overview of RL and Transfer Learning (TL), highlighting their fundamental concepts and applications in mobility control. Sec. III delves into the specifics of TRL, discussing its methodologies and objectives in enhancing learning efficiency and performance across different tasks or domains. Sec. V analyzes the recent trends and applications of TRL in mobility control. Sec. VI concludes the paper by summarizing the key findings in the field of TRL for vehicle control.

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on April 08,2026 at 05:43:12 UTC from IEEE Xplore. Restrictions apply.

II. Background

A. Reinforcement Learning

RL is a methodology in which an agent learns the optimal action policy by interacting with its environment. The agent selects an action from a given state, receives a reward from the environment as a result, and then transitions to the next state. The goal of the agent is to maximize the long-term cumulative reward. RL is modeled as a Markov Decision Process (MDP), which provides the mathematical framework for this learning paradigm. An MDP is characterized by the following components

- **State Space (S):** The set of all possible states that the agent can occupy.
- **Action Space (A):** : The set of all possible actions the agent can take.
- **Transition Probability (P):** The probability $P(s' \mid s, a)$ that action a in state s will lead to state s'.
- **Reward Function (R):** The immediate reward received after transitioning from state s to state s' due to action a.
- **Discount Factor (γ):** A factor $\gamma \in [0, 1]$ that quantifies the agent's preference for immediate rewards over future rewards.

B. Transfer Learning

TL is a concept that has been extensively studied in the field of machine learning. It is a methodology that aims to improve performance by applying knowledge learned from one task to related tasks. Fig. 1 illustrates the advantages of TL. Notably, it highlights the "jump start" property, where a model trained in the source domain exhibits high initial performance in the target domain. Additionally, TL can lead to higher overall performance compared to conventional learning methods. The primary goal of TL is to enable learners to apply the knowledge, experience, or representations gained from learning a specific task to a new task, thereby allowing the learning process to proceed more quickly and efficiently than if starting from scratch. In deep learning, TL is frequently employed to address problems with limited data. This is particularly important because deep learning models generally require large amounts of data for training, which can be difficult or costly to obtain.

1) **To save time and resources**: Training deep learning models from scratch requires significant computational power and time. However, by leveraging TL, the computational requirements and time needed to achieve the desired performance can be significantly reduced.
2) **To enhance model performance**: TL allows a model to retain the features and knowledge obtained from prior tasks. As a result, when applied to similar new tasks, it can effectively improve performance.
3) **To enable the solution of tasks with limited data**: A pre-trained model can be utilized to solve tasks with limited data by applying the knowledge it has already acquired about the features of the data.

C. Mobility Control

Mobility Control refers to the methodologies and systems used to manage and direct the movement of various vehicles and platforms, including unmanned aerial vehicle (UAV), autonomous ground Vehicle (AGV), autonomous underwater vehicles (AUV), and spacecraft. Effective mobility control is crucial for ensuring stability, safety, and efficiency of these systems, especially in dynamic and complex environments. This concept encompasses various aspects, including control systems, communication technologies, energy management, and transportation infrastructure. Mobility Control encompasses the management of vehicle dynamics, communication systems, energy management, and integration with transportation infrastructure. It ensures the safe, efficient, and adaptive operation of various mobile platforms in dynamic environments.

III. Transfer Learning on Reinforcement Learning

TRL is an advanced approach that combines the principles of TL and RL to improve learning efficiency and performance in new tasks or domains. TRL addresses these challenges by leveraging knowledge gained from a source domain M_s and transferring it to a target domain M_t, where M_s and M_t represent different Markov Decision Processes.

- **Source Domain (M_s)**: The source domain is where the agent initially learns. It is defined by the tuple $M_s = \{S_s, A_s, P_s, R_s\}$
- **Target Domain (M_t)**: The target domain is the new environment or task where the agent will apply the transferred knowledge. It is defined by the tuple $M_t = \{S_t, A_t, P_t, R_t\}$

TRL addresses these challenges by leveraging knowledge gained from M_s and transferring it to M_t, where M_s and M_t represent different Markov Decision Processes (MDPs). The primary objective of TRL is to efficiently find the optimal policy pi_t^* in M_t by leveraging the knowledge from M_s. Mathematically, this objective can be expressed as:

$$\pi_t^* = \arg\max_{\pi} \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t R_t(s_t, a_t) \mid M_t, \pi\right] \tag{1}$$

where π, $R_t(s_t, a_t)$ and $\mathbb{E}$ represent the policy in the target domain M_t, the reward received at time step t in the target domain M_t and the expected value under the policy π. The aim is to find pi_t^* that maximizes the expected cumulative reward in the target domain, utilizing the transferred knowledge from the source domain to accelerate and improve the learning process.

IV. Mobility Control Application and algorithms

A. Mobility Application

Research on transferring the policies of autonomous vehicles trained in intersection traffic control simulations to more complex domains has been actively conducted. For instance, studies have explored the feasibility of applying models trained in one driving task to different intersection scenarios. These

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on April 08,2026 at 05:43:12 UTC from IEEE Xplore. Restrictions apply.

studies have demonstrated a success rate of over 90% when the scenarios are similar, and even a 70% success rate in dissimilar scenarios, indicating an improvement in learning efficiency [11]. Additionally, research has been conducted on MARL utilizing TL [12]. Specifically, adversarial learning, which employs adversarial agents to disrupt the inputs and outputs of autonomous vehicles, has been applied to improve this process, aiming to implement autonomous systems that reduce traffic congestion and increase energy efficiency [13]. For this purpose, a simple simulation environment based on MARL was developed using a zero-shot transfer method and then transferred to a scaled smart city testbed environment [14]. Moreover, a method was proposed to address the problem of variable speed control using RL strategies through TL algorithms. This study demonstrated that models trained in one scenario could be transferred to other similar scenarios, thereby shortening the learning process while maintaining or improving performance. Meta-RL, which allows rapid adaptation to unseen tasks based on prior knowledge, was utilized to investigate whether an Autonomous Mobility-on-Demand (AMoD) system, operated by autonomous vehicles, could adapt to the data from cities with limited datasets using models trained on data from cities with sufficient data [15]. Furthermore, research addressing the method by which UAVs (Unmanned Aerial Vehicles) track maneuvering targets in uncertain environments has proposed an efficient path planning method for UAV maneuvering target tracking and obstacle avoidance using TRL techniques. This study first trained the UAVs in a task involving straight-line target tracking and then adapted them to environments with arbitrary complex paths, moving targets, and obstacles [16]. In situations where UAVs must consider battery life and recharging, research was also conducted to optimize the age of information, battery usage, flight paths, and transmission scheduling jointly in new complex environments through TL techniques [17]. Additionally, research has been conducted on combat scenarios involving multiple UAVs, where the policies trained with three UAVs in a specific environment were subsequently applied to a combat scenario involving five agents [18]. Similarly, studies have employed TL in MARL involving homogeneous robotic mobile agents [19]. Research has also been conducted to ensure models trained in source domains with adversarial UAV agents and sensing errors could operate effectively in a dual-UAV transportation scenario [20]. To address the challenges that arise when multiple unmanned surface vehicles (Multi-USVs) cooperate in complex tasks such as ocean observation data collection, a TL-based deep reinforcement learning (DRL) is proposed. This approach involves learning optimal policies in an environment where strategies like task allocation and collision avoidance are employed within a small group, and then applying these policies to new groups. As a result, tasks can be performed efficiently from the initial stages, leading to reduced training time and enhanced performance compared to scenarios where TL is not utilized [21]. TL has also been applied to the control algorithms of heterogeneous agents. By integrating Gaussian mixture models with TL techniques, studies have focused on enabling effective operation across agents with different reward structures and action spaces [22].

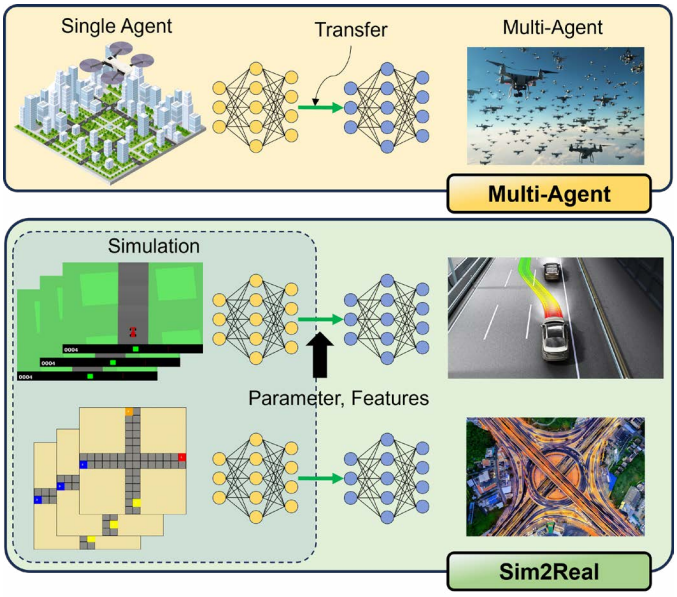

Fig. 2: Concept of Transfer Learning on Reinforcement learning.

B. Application for Sim2Real

In the context of Sim2Real implementations of RL applicable to real-world scenarios, TL techniques have been extensively studied [23]. Fig. 2 provides an overview of the areas where TL has been applied to Reinforcement Learning. Research has shown that policies learned in lane-keeping and overtaking simulations were effectively transferred to real-agent environments [24]. Additionally, studies have addressed the Sim2Real transfer of DRL controllers for nonlinear system control of tower cranes. These controllers were developed to position the trolley at a target location while simultaneously suppressing load oscillation, with the research being applied to a real-world tower crane model to optimize learning performance [25].

V. Challenges and Future Directions

In this section, we analyze the trends in research papers that apply TL techniques to RL for vehicle control. We discuss the current limitations of these approaches and suggest potential directions for future research.

1) **Sim2Real Transfer and Real-World Adaptation:** The Sim2Real gap presents significant challenges in transferring RL policies from simulation to real-world environments due to discrepancies in dynamics and unanticipated variables. Advanced domain randomization and feedback mechanisms, alongside online learning, will help refine and adapt policies for real-world applications.
2) **Multi-Agent Collaboration and Coordination:** Effective collaboration among multiple agents, particularly in autonomous vehicle systems, remains a challenge due to potential conflicts and suboptimal outcomes from independent learning. Enhancing MARL through improved

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on April 08,2026 at 05:43:12 UTC from IEEE Xplore. Restrictions apply.

collaboration strategies and value decomposition networks is crucial, focusing on reliability and coordinated actions in complex environments.

3) **Scalability and Real-Time Application:** Scaling RL-based control systems for real-time applications poses challenges in achieving the necessary speed and efficiency. Research should focus on improving RL algorithm efficiency, leveraging hardware acceleration, and exploring distributed learning to meet real-time demands.

VI. Concluding Remarks

In this paper, we have comprehensively analyzed to RL for vehicle control, particularly in autonomous driving and unmanned systems. The integration of TL with RL has demonstrated significant potential in addressing challenges related to Sim2Real transfer, multi-agent collaboration, and scalability in real-time applications. Despite the promising results, several challenges remain unresolved. The Sim2Real gap continues to hinder the seamless application of policies learned in simulation to real-world environments. Similarly, ensuring effective collaboration and coordination among multiple agents in complex, dynamic environments is an ongoing challenge. Additionally, scaling RL-based control systems to meet the demands of real-time applications requires further research and innovation.

Acknowledgement

This research was supported by the MSIT (Ministry of Science and ICT), Korea, under the ITRC (Information Technology Research Center) support program (IITP-2024-RS-2024-00436887) supervised by the IITP (Institute for Information & Communications Technology Planning & Evaluation). J. Kim is the corresponding author of this paper(joongheon@korea.ac.kr).

References

[1] C. Ding, I. W.-H. Ho, E. Chung, and T. Fan, "V2X and deep reinforcement learning-aided mobility-aware lane changing for emergency vehicle preemption in connected autonomous transport systems," *IEEE Transactions on Intelligent Transportation Systems*, vol. 25, no. 7, pp. 7281–7293, January 2024.

[2] H. Taghavifar, C. Wei, and L. Taghavifar, "Socially intelligent reinforcement learning for optimal automated vehicle control in traffic scenarios," *IEEE Transactions on Automation Science and Engineering*, pp. 1–12, January 2024.

[3] Y. Fu, C. Li, F. R. Yu, T. H. Luan, and Y. Zhang, "A decision-making strategy for vehicle autonomous braking in emergency via deep reinforcement learning," *IEEE Transactions on Vehicular Technology*, vol. 69, no. 6, April 2020.

[4] H. Unjhawala, T. Hansen, H. Zhang, S. Caldraru, S. Chatterjee, L. Bakke, J. Wu, R. Serban, and D. Negrut, "An expeditious and expressive vehicle dynamics model for applications in controls and reinforcement learning," *IEEE Access*, vol. 12, pp. 33 000–33 015, February 2024.

[5] F. Zhang, Q. Yang, and D. An, "CDDPG: A deep-reinforcement-learning-based approach for electric vehicle charging control," *IEEE Internet of Things Journal*, vol. 8, no. 5, pp. 3075–3087, August 2021.

[6] M. Choi, T. Xiang, and J. Kim, "Intelligent caching for seamless high-quality streaming in vehicular networks: A multi-agent reinforcement learning approach," *IEEE Transactions on Intelligent Vehicles*, vol. 9, no. 2, pp. 3672–3686, February 2024.

[7] C. Park, G. S. Kim, S. Park, S. Jung, and J. Kim, "Multi-agent reinforcement learning for cooperative air transportation services in city-wide autonomous urban air mobility," *IEEE Transactions on Intelligent Vehicles*, vol. 8, no. 8, pp. 4016–4030, August 2023.

[8] M. Shin, D.-H. Choi, and J. Kim, "Cooperative management for pv/ess-enabled electric vehicle charging stations: A multiagent deep reinforcement learning approach," *IEEE Transactions on Industrial Informatics*, vol. 16, no. 5, pp. 3493–3503, 2020.

[9] Z. Chu, F. Wang, T. Lei, and C. Luo, "Path planning based on deep reinforcement learning for autonomous underwater vehicles under ocean current disturbance," *IEEE Transactions on Intelligent Vehicles*, vol. 8, no. 1, pp. 108–120, February 2023.

[10] W. J. Yun, J. Park, and J. Kim, "Quantum multi-agent meta reinforcement learning," in *Proc. AAAI Conference on Artificial Intelligence*, vol. 37, no. 9, February 2023, pp. 11 087–11 095.

[11] H. Shu, T. Liu, X. Mu, and D. Cao, "Driving tasks transfer using deep reinforcement learning for decision-making of autonomous vehicles in unsignalized intersection," *IEEE Transactions on Vehicular Technology*, vol. 71, no. 1, pp. 41–52, October 2022.

[12] H. Ge, D. Gao, L. Sun, Y. Hou, C. Yu, Y. Wang, and G. Tan, "Multi-agent transfer reinforcement learning with multi-view encoder for adaptive traffic signal control," *IEEE Transactions on Intelligent Transportation Systems*, vol. 23, no. 8, pp. 12 572–12 587, October 2022.

[13] Z. Yao, S. Xia, Y. Li, G. Wu, and L. Zuo, "Transfer learning with spatial–temporal graph convolutional network for traffic prediction," *IEEE Transactions on Intelligent Transportation Systems*, vol. 24, no. 8, pp. 8592–8605, March 2023.

[14] Z. Xu, C. Tang, and M. Tomizuka, "Zero-shot deep reinforcement learning driving policy transfer for autonomous vehicles based on robust control," in *Proc. International Conference on Intelligent Transportation Systems (ITSC)*, Maui, Hawaii, USA, November 2018, pp. 2865–2871.

[15] D. Gammelli, K. Yang, J. Harrison, F. Rodrigues, F. Pereira, and M. Pavone, "Graph meta-reinforcement learning for transferable autonomous mobility-on-demand." New York, NY, USA: Association for Computing Machinery, August 2022, p. 2913–2923.

[16] B. Li, Z. peng Yang, D. qing Chen, S. yang Liang, and H. Ma, "Maneuvering target tracking of UAV based on MN-DDPG and transfer learning," *Defence Technology*, vol. 17, no. 2, pp. 457–466, April 2021.

[17] M. Yi, X. Wang, J. Liu, Y. Zhang, and R. Hou, "Multitask transfer deep reinforcement learning for timely data collection in rechargeable-UAV-aided IoT networks," *IEEE Internet of Things Journal*, vol. 10, no. 23, pp. 20 545–20 559, August 2023.

[18] D. Xu, P. Qiao, and Y. Dou, "Aggregation transfer learning for multi-agent reinforcement learning," in *Proc. International Conference on Big Data & Artificial Intelligence & Software Engineering (ICBASE)*, Zhuhai, China, September 2021, pp. 547–551.

[19] S. Wen, Z. Wen, D. Zhang, H. Zhang, and T. Wang, "A multi-robot path-planning algorithm for autonomous navigation using meta-reinforcement learning based on transfer learning," *Applied Soft Computing*, vol. 110, p. 107605, October 2021.

[20] S. Chen, G. Liu, Z. Zhou, K. Zhang, and J. Wang, "Robust multi-agent reinforcement learning method based on adversarial domain randomization for real-world dual-UAV cooperation," *IEEE Transactions on Intelligent Vehicles*, vol. 9, no. 1, pp. 1615–1627, August 2024.

[21] J. Zhang, J. Ren, Y. Cui, D. Fu, and J. Cong, "Multi-USV task planning method based on improved deep reinforcement learning," *IEEE Internet of Things Journal*, vol. 11, no. 10, pp. 18 549–18 567, August 2024.

[22] H. Shi, J. Li, J. Mao, and K.-S. Hwang, "Lateral transfer learning for multiagent reinforcement learning," *IEEE Transactions on Cybernetics*, vol. 53, no. 3, pp. 1699–1711, September 2023.

[23] M. Kaspar, J. D. Muñoz Osorio, and J. Bock, "Sim2Real transfer for reinforcement learning without dynamics randomization," in *Proc. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, October 2020, pp. 4383–4388.

[24] D. Li and O. Okhrin, "A platform-agnostic deep reinforcement learning framework for effective Sim2Real transfer in autonomous driving," April 2024. [Online]. Available: https://arxiv.org/abs/2304.08235

[25] M. B. Mohiuddin, A. G. Haddad, I. Boiko, and Y. Zweiri, "Zero-shot sim2real transfer of deep reinforcement learning controller for tower crane system," *IFAC-PapersOnLine*, vol. 56, no. 2, pp. 10 016–10 020, July 2023.

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on April 08,2026 at 05:43:12 UTC from IEEE Xplore. Restrictions apply.